

TARGET

Use the definition of matter and a particle model to make and limit claims about a sample.

ENGAGE

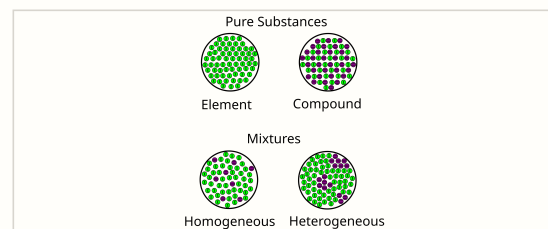
Air is invisible, light is visible, and sound can be heard. Which of the three is matter, and what evidence should decide?

TEACH

- Use mass and occupied space—not visibility—as the classroom test for matter.
- A particle symbol represents a type of particle; repeated symbols can reveal composition and distribution.
- Identical particles support a pure-substance claim, while more than one particle type supports a mixture claim.
- Models are not to scale and do not prove every property of the real sample.

TERMS

matter: anything that has mass and occupies space
 macroscopic: the scale of observations made directly or with ordinary tools
 particle model: a simplified representation of particles and their arrangement
 model limitation: a feature of reality that a representation does not show accurately



Particle types and distribution provide model evidence.

WORKED EXAMPLE

A sealed diagram contains identical two-atom particles. What can be concluded, and what remains unknown?

1. Every represented particle is identical, so the model supports a pure-substance claim.
2. Inspect the atom symbols inside one particle: the same element would indicate an element; different elements would indicate a compound.
3. If the symbols do not distinguish atom identity, the model lacks the evidence needed for the second decision.

RESULT: The sample is modeled as pure. Element versus compound cannot be decided without evidence about the atom types in each particle.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE**L1-1. Classify air in a sealed syringe, visible light, and sound as matter or not matter. Defend each decision.**

Scaffold: Apply both parts of the definition: mass and occupied space.

INDEPENDENT**L1-2. In the particle visual, what evidence distinguishes a homogeneous mixture from a heterogeneous mixture?**

INDEPENDENT**L1-3. State one useful feature and one limitation of the particle visual.**

INDEPENDENT**L1-4. A sample expands to fill its container. Explain why this observation alone does not prove it is not matter.**

EXIT TICKET**L1-5. What additional evidence is needed to decide whether identical two-atom particles represent an element or a compound?**

Learn and model

Name: _____ Date: _____ Period: _____

TARGET**Classify elements, compounds, homogeneous mixtures, and heterogeneous mixtures from composition evidence.****ENGAGE**

A clear saltwater sample looks uniform. Does uniform appearance prove that it is a pure substance?

TEACH

- First decide pure substance versus mixture; then choose the more specific label.
- An element contains one element type; a compound contains different elements chemically combined in a fixed composition.
- A homogeneous mixture is uniform at the sampled scale but still contains multiple substances.
- A heterogeneous mixture has nonuniform composition or visibly distinct regions. Physical separation evidence supports a mixture claim.

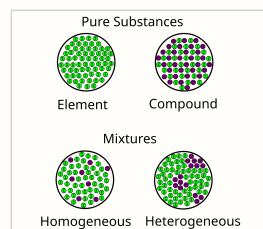
TERMS

pure substance: matter with constant composition and characteristic properties

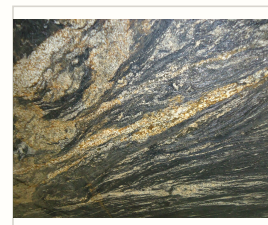
element: a pure substance made of one chemical element

compound: a pure substance containing elements chemically combined in fixed proportions

mixture: a physical combination of substances with variable composition



Follow composition first, then distribution.



Visible mineral regions make granite nonuniform at this scale.

WORKED EXAMPLE**A clear saltwater sample leaves salt after the water is removed by a physical process. Classify the original sample.**

1. The sample appears uniform, so it is homogeneous at the observed scale.
2. A physical process recovers components, which supports the presence of more than one substance.
3. Uniform distribution plus multiple substances identifies a homogeneous mixture.

RESULT: Saltwater is a homogeneous mixture, not a pure compound.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L2-1. Classify a clean copper wire as element, compound, homogeneous mixture, or heterogeneous mixture. Explain.

Scaffold: Ask whether the sample contains one element type or more than one substance.

INDEPENDENT

L2-2. Classify distilled water and explain why it is not an element.

INDEPENDENT

L2-3. Classify clean, well-mixed air at one location. Explain why uniform does not mean pure.

INDEPENDENT

L2-4. Classify granite with visibly different mineral regions.

EXIT TICKET

L2-5. A model has two particle types evenly intermingled. Give the most specific label and two pieces of evidence.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Classify physical and chemical properties and distinguish intensive from extensive properties.

ENGAGE

Copper conducts electricity and can react with oxygen. Which statement can be checked without changing copper into a new substance?

TEACH

- Color, density, melting point, and conductivity are physical properties when measured without changing identity.
- Flammability and reactivity describe possible chemical behavior, even when no reaction is occurring at that moment.
- Mass and volume are extensive because they change with sample amount.
- Density and melting point are intensive under the same stated conditions and can help identify a substance.

TERMS

physical property: a characteristic observed or measured without changing identity

chemical property: the ability of a substance to undergo a chemical change

intensive property: a property independent of sample amount under stated conditions

extensive property: a property that depends on the amount of sample



Color and luster are observable physical properties of copper.



Flammability describes the ability to burn; the property exists before ignition.

WORKED EXAMPLE

Classify density and mass as physical or chemical and as intensive or extensive.

1. Both can be measured without changing chemical identity, so both are physical properties.
2. Density remains characteristic when sample amount changes under the same conditions, so it is intensive.
3. Mass increases or decreases with sample amount, so it is extensive.

RESULT: Density is physical and intensive; mass is physical and extensive.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L3-1. Classify melting point as physical or chemical and intensive or extensive.

Scaffold: Ask whether identity changes during measurement and whether the value depends on sample amount.

INDEPENDENT

L3-2. Classify volume as physical or chemical and intensive or extensive.

INDEPENDENT

L3-3. Classify flammability and explain why it is not a physical property.

INDEPENDENT

L3-4. A lab card says 'Copper conducts electric current.' Is this a property or a change? Classify it.

EXIT TICKET

L3-5. Why is density often more useful for identification than volume?

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Classify changes by the new-substance criterion and evaluate common clues in context.

ENGAGE

Bubbles appear when a cold soda is opened and when two reacting solutions are mixed. Do bubbles prove the same kind of change in both cases?

TEACH

- Melting, freezing, crushing, dissolving in a recoverable context, and physical separation do not by themselves require a new substance.
- Burning, decomposition, and reactions that form products with new compositions are chemical changes.
- A clue must be compared with alternatives: bubbles may be a gas product, boiling, or released dissolved gas.
- An open container may lose gaseous products from measurement; this does not violate conservation of matter.

TERMS

physical change: a change in form or state that does not require a new substance

chemical change: a process in which atoms rearrange to form new substances

evidence: an observation or result used to support or challenge a claim

conservation of matter: matter is not created or destroyed in a closed-system chemical change



Melting changes state while the substance remains water.



Burning forms new substances and is chemical.

WORKED EXAMPLE

Salt dissolves in water and is later recovered by evaporating the water. Classify the process in this context.

1. Dissolving produces a uniform solution but does not by itself establish a new substance.
2. The original salt is recovered by a physical method.
3. Recovery supports a physical-change/separation claim for this classroom context.

RESULT: Physical change and physical separation in this context.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L4-1. Ice melts in a sealed bag. Classify the change and identify what remains the same.

Scaffold: Ask whether the substance's chemical identity changes.

INDEPENDENT

L4-2. Candle wax burns in oxygen and produces different substances. Classify the change and cite decisive evidence.

INDEPENDENT

L4-3. A tablet is crushed into powder without reacting. Classify the change.

INDEPENDENT

L4-4. Explain why bubbles support but do not by themselves prove a chemical-change claim.

EXIT TICKET

L4-5. A solid is strongly heated in an open container and a gas escapes. Why can the measured solid mass decrease without violating conservation?

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Choose and sequence physical separation methods from differences in particle size, solubility, volatility, magnetism, or phase behavior.

ENGAGE

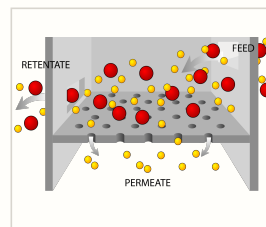
A beaker contains iron filings, salt, sand, and water. Which separation should happen first, and why does order matter?

TEACH

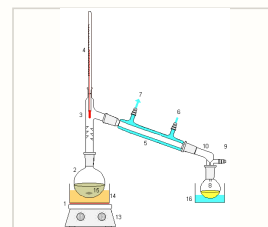
- Use magnetism before adding water when a dry magnetic component is present.
- Use filtration for an insoluble solid in a liquid; dissolved solute passes with the liquid.
- Use distillation when the goal includes collecting the more volatile liquid as well as the residue.
- Use chromatography to test whether a soluble colored sample contains components with different attractions and travel rates.

TERMS

filtration: separation using particle size and insolubility
 distillation: separation using differences in volatility or boiling behavior
 chromatography: separation using different attractions and travel rates
 decantation: separation by carefully pouring one phase from another



Residue remains on the filter; filtrate passes through.



Vapor is condensed and collected during simple distillation.

WORKED EXAMPLE

A uniform blue liquid yields colorless liquid and blue solid after distillation. What claim is supported?

1. The original sample appears homogeneous, but appearance alone cannot establish purity.
2. Distillation is a physical method based on volatility or boiling behavior.
3. Recovering two components by a physical process supports a mixture claim.

RESULT: The original sample is a homogeneous mixture with at least two components.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE**L5-1. Choose a method to separate insoluble sand from water and name the property difference.**

Scaffold: Decide whether the solid is dissolved and whether its particles can be retained.

INDEPENDENT**L5-2. A saltwater sample must yield both water and salt. Choose a method and explain why evaporation alone is incomplete.**

INDEPENDENT**L5-3. Choose a method to test whether black marker ink contains more than one dye.**

INDEPENDENT**L5-4. Choose the first method for a dry mixture of iron filings and table salt.**

EXIT TICKET**L5-5. Explain why separating a mixture is usually physical even when several steps are required.**



Cumulative Mixed Review

STUDENT REVIEW

Use ideas from every lesson

Name: _____ Date: _____ Period: _____

R1. Explain why air is matter even though it is usually invisible.

R2. A particle model shows one repeated particle made of two different atom types. Classify the sample.

R3. Classify salad dressing with visible oil and water layers.

R4. Classify electrical conductivity and flammability as physical or chemical properties.

R5. Explain why mass is extensive but density is intensive.



Cumulative Mixed Review

Use ideas from every lesson

Name: _____ Date: _____ Period: _____

R6. Classify boiling water and burning paper as physical or chemical changes.

R7. Bubbles form when a liquid is warmed. What additional evidence is needed before claiming a chemical change?

R8. Choose a method for oil and water layers and name the property evidence.

R9. A distillation starts with 120.0 g and recovers 92.0 g liquid plus 27.6 g solid. Calculate percent recovery.

R10. Why can a uniform appearance support but not prove a homogeneous-mixture classification?
